The Ultimate Computer, Ever. Version 1

The TUCE-V1 is a hardware Brainfuck interpreter.

TUCE Registers and Flags

Abbreviation	Full Name	Size
PC	Program Counter	16 Bits
MP	Memory Pointer	16 Bits
DT	Depth Tracker	16 Bits
INR	Input Register	8 Bits
OUTR	Output Register	8 Bits
IR	Instruction Register	8 Bits
MIR	Memory Input Register	8 Bits
MOR	Memory Output Register	8 Bits
OWF	Output Write Flag	1 Bit
MOWF	Memory Output Write Flag	1 Bit
FLF	Forward Loop Flag	1 Bit
BLF	Backward Loop Flag	1 Bit
IDU	Increment Decrement Unit	8 Bits
SC	Stage Counter	2 Bits

Other Abbreviations

Abbreviation	Full Name
CLK	Clock Input
RST	Reset
BF	Brainfuck
LM	Loop Mode
NM	Normal Mode
IO	Input / Output

TUCE Modes

Mode	Description
Normal	TUCE will interpret instructions as normal.
Loop	TUCE will ignore any instruction that isn't `[` or `]`

Brainfuck Instructions

Instruction	Function	Binary Value
`N`	Do nothing	00000000
`>`	Increment the MP by one	00000001
`<`	Decrement the MP by one	00000010
`+`	Increment the current cell by one	00000100
`-`	Decrement the current cell by one	00001000
`.`	Output the current cell	00010000
`,`	Input to the current cell	00100000
`[`	If the current cell is zero, move `PC` forwards to the instruction after the matching `]`	01000000
`]`	If the current cell is non-zero, move `PC` backwards to the instruction after the matching `[`	10000000

- Please note that `N` is not a real BF instruction, and only exists on the TUCE because of the way instructions are decoded.
- Instructions with values other than described above will produce undefined results.

TUCE Stages and Operation

- Rise refers to when `CLK` goes high.
- Fall refers to when `CLK` goes low.
- SC increments every time `CLK` goes low.

S1 – Initialize (Both Modes)

Rise

- Load `IR`

Fall

Instruction	Operation
`>`	Increment `MP`
`<`	Decrement `MP`
`+` `-` `.` `[` `]`	Load `MIR`
`,`	Load `INR`

TUCE Stages and Operation: Normal Mode

S2 – Memory / IO

Rise

Instruction	Operation
`[`	Set `FLF` if `MIR` is zero
`]`	Set `BLF` if `MIR` is non-zero

Fall

Instruction	Operation
`+` `-`	Load `MOR` with `IDU`
`.`	Load `OUTR` with `MIR`
`,`	Load `MOR` with `INR`
`+` `-` `,`	Set `MOWF`
`.`	Set `OWF`

S3 – Delay

- This does nothing but waste a cycle.
- It does this so memory and IO devices have a clock cycle to use the data that was output.

TUCE Stages and Operation: Loop Mode

S2 - Instruction Checking

Rise

- Nothing.

Fall

Flag	Operation
`FLF`	If `IR` is a matching `]`, decrement `DT`; otherwise, increment `DT`.
`BLF`	If `IR` is a matching `[`, decrement `DT`; otherwise, increment `DT`.

S3 - Depth Checking

Rise

- If `DT` is equal to zero, clear loop flags.

Fall

- Nothing.

S4 – Finalize (Both Modes)

Rise

Mode / Flag	Operation
`NM` `FLF`	Increment `PC`
`BLF`	Decrement `PC`
`NM` `FLF` `BLF`	Clear `MOWF`, `OWF`, `OUTR`, `MOR`, `MIR`, `INR`, and `IR`

Fall

- Nothing.